



With thanks to
CASIO

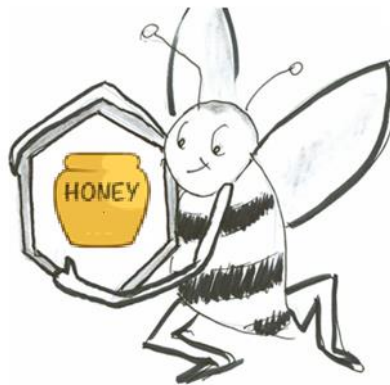
TOUR DE MATHS 2022

LEG 3

12 May 2022

Jeppe High School For Girls

QUESTION PAPER



A study has shown that honeybees can add and subtract if trained to do so. This shows that animals do not need a huge brain, or anything resembling human language to think in terms of numbers.



Questions worth 5 honey pots each

Question 1

Five bees were racing to the same flower.

A got to the flower before B, but after C.

D got to the flower before E, but after B.

What was the finishing order from first to last?



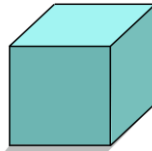
Question 2

If $a \text{ bee } b$ means a^b and $a \text{ bee } b$ means $\sqrt[b]{a}$ then what is the value of:

$$\left((2 \text{ bee } 6) \text{ bee } 3 \right) \text{ bee } 2$$

Question 3

How many cubes with edges of length 2cm can be made from a rectangular block 6cm long, 4cm wide and 2cm high?



Question 4

The value of $2222^2 - 2221^2$ is...

Question 5

How many two-digit numbers, less than 50, are exactly 4 times the sum of their digits?

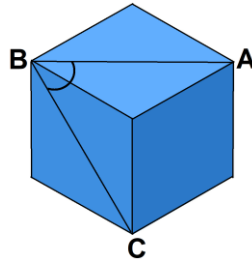
Question 6

Increasing $800 - 10y$ by 10% gives $600 - 6y$.

What is the value of y ?

Question 7

In the cube, how big is angle $\widehat{ABC}$?



Question 8

You are allowed to perform **TWO** operations with numbers: **doubling**, and **increasing by 1**. If you start from zero, what is the smallest number of operations needed to build up the number 100?

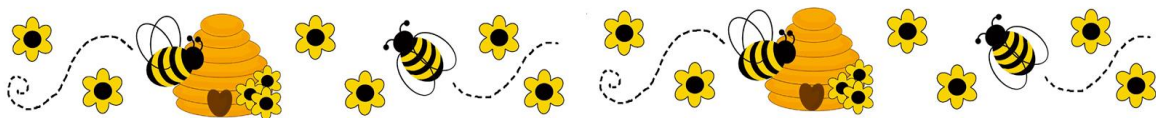
Question 9

Three normal dice are rolled. What is the probability that they all show the same number?



Question 10

25 pots of honey are distributed in 14 boxes. Each box contains 1, 2 or 3 honey pots. 7 boxes contain just one honey pot each. How many boxes will have only 2 honey pots in each?





Questions worth 10 honey pots each

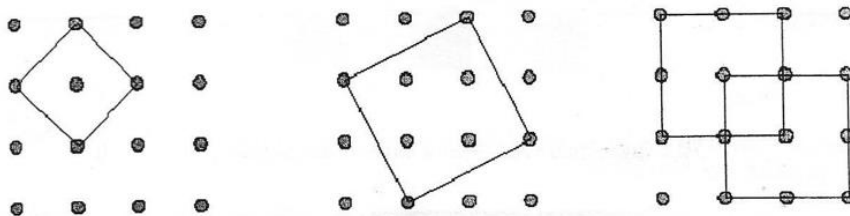
Question 11

Layla has 10 beehives. She arranges the hives in the shape of a _____ in order to have 5 lines with 4 hives in each line (the lines may intersect).



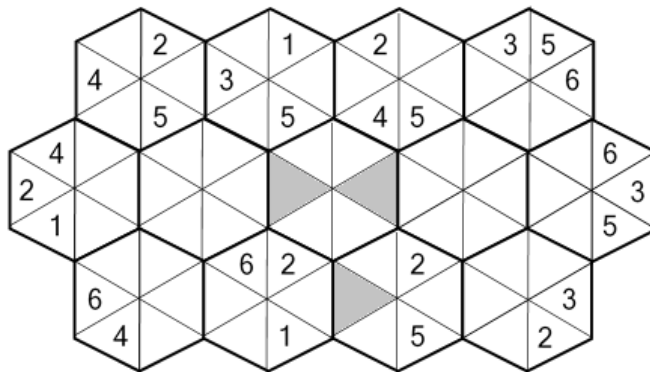
Question 12

By joining dots together, how many squares can be drawn on a grid measuring 4 dots by 4 dots? Some examples have been provided below:



Question 13

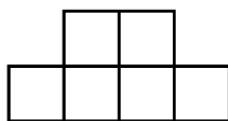
Fit the numbers 1 – 6 in each hexagon. Where the sides of different hexagons meet, the adjoining segments will have the same number value. No number can be repeated in a hexagon.



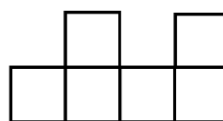
The sum of the numbers in the **shaded** triangles is: _____

Question 14

What is the least number of cubes needed to make the arrangement shown below:



(a) Front view



(b) side view

Question 15

Use the clues to crack the code to determine the number of bees in a hive:

2	6	5
3	8	7
2	3	4
4	7	1
?	?	?



One number is correct and well placed



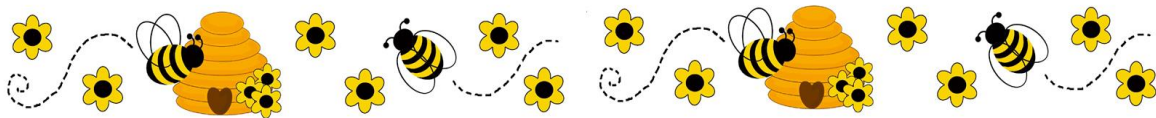
Nothing is correct



One number is correct but wrongly placed



Two numbers are correct but wrongly placed



Questions worth 20 honey pots each

Question 16

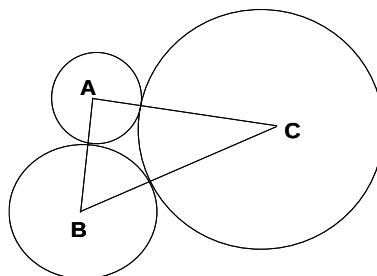
The natural numbers are arranged in the pattern below. In which of the following rows does 2022 lie?

Row 1			4			12				20		
Row 2		2		7	10		15		18		23	
Row 3	1		5		9	13		17		21		25
Row 4		3		8	11		16		19		24	
Row 5			6			14				22		

Question 17

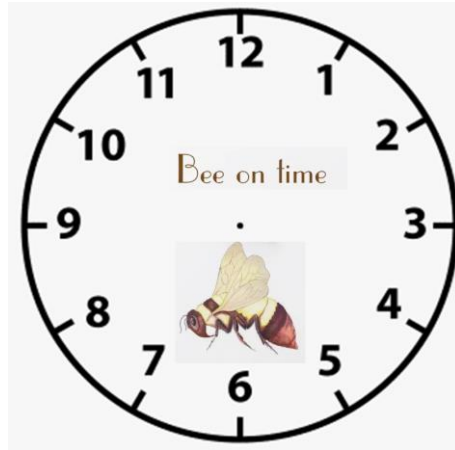
The vertices of a right-angled triangle are the centres of three mutually, externally tangential circles as shown. $AB = 3$; $AC = 4$; $BC = 5$. What is the sum of the areas of these circles?

Leave your answer in terms of π .



Question 18

What is the angle between the minute hand and the hour hand at quarter past three?

**Question 19**

Stranded on a deserted island, Sam and his crew are left with only a 40 litre container of milk. To conserve the milk they decide that on the first day they will drink one litre of milk and then refill the container back up with water. On the 2nd day they will drink 2 litres and refill the container. On the 3rd day they will drink 3 litres and so on... By the time all the milk is finished, how much water have they drunk altogether?

**Question 20**

A faulty car odometer proceeds from digit 3 to digit 5, always skipping the digit 4, regardless of position. For example, the odometer changes immediately from 000039 to 000050. If the odometer now reads 002022, how many kilometres has the car actually travelled?

